`problems/MHD`: DynSGS and positivity across the nine cases

Summary of the per-case notes, as coded on branch `sm/MHD` (commit `dcc3653`)

How to read the underlines. Solid underline: the setting (or the code path) is *identical in every case of `problems/MHD` to which it applies* — all nine cases for general settings, the eight `DSGS_MHD` cases for DynSGS settings, the three `:positivity_method => "conservative"` cases for the element limiter. Dashed underline: shared by a *majority* of those cases but not all of them. No underline: specific to this case or to a minority. The cross-case comparison is in `summary.pdf` of this directory.

This directory holds one note per case (`<case>.tex`/`.pdf`) and this summary. Every value was read from the case's deck and files and from the kernels they reach on `sm/MHD`; where the deck is silent the value is `mod_inputs.jl`'s default, and environment overrides (`JEXPRESSO_BW_*`, `_SV_*`, `_OT_*`) are taken unset. Column keys:

BW	`brioWu1d`	KH	`kelvinHelmholtzChan2022`	ROT	`rotorDaoNazarov2022`
SV	`smoothVortex`	FE	`fluxEmergenceSon2025`	BL	`blastBalsaraSpicer1999`
OT	`orszagTangBormanis2024`	FED	`fluxEmergenceSon2025DSGS`	AJ	`astroJetWuShu2018`

In one paragraph

Eight of the nine cases stabilize the collocated CG–SEM with the residual-based `DSGS_MHD` viscosity; **KH uses Smagorinsky** and no DynSGS key. All eight share the element-wise coefficient $\nu_K = \max\big(C_{\min}\Delta_K w_K,\ \min(C_{\max}\Delta_K w_K,\ C_R\Delta_K^2\mathcal{R}_K)\big)$ with $\Delta_K = \Delta_K^{elem}/(N+1)$, $w_K = \max(|\mathbf{v}| + c_f)$, the max of the normalized residual over the eight non-ψ slots, mass diffusion on, Pr = 1, `:dsgs_rel`= 1, no cutoff, no freezing; they split on the sensor (`legacy` 5, `residual` 3), the normalization scope (`element` 5, `domain` 3), the operator (a Laplacian of the conserved state or of its departure in 6, the physical form in OT and SV — SV since `f9cdb02`, a hybrid of the two before), and the hold (2 steps in 6, none in BL, AJ). Positivity is enforced by a *state* modification in only four cases: the conservative element limiter (ROT, BL, AJ) and a non-conservative case-local floor (FE); BW, FE, FED and AJ additionally floor p (and AJ ρ) inside the flux only. FED is the limiter-free twin of FE.

Table 1 — Problem, discretization, time integration

	BW	SV	OT	KH	FE	FED	ROT	BL	AJ
Dim. / slots	1D / 8	2D / 9	2D / 9	2D / 9	2D / 9	2D / 9	2D / 9	2D / 9	2D / 9
γ	2	5/3	5/3	5/3	1.05	1.05	1.4	1.4	1.4
Domain	$(0, 1)$	$[-10, 10]^2$	$[0, 1]^2$	periodic square (external mesh)	$80 \times 35\, H_0$	$80 \times 35\, H_0$	$[0, 1]^2$	$[0, 1]^2$	$[0, \frac{1}{2}] \times [0, 1.5]$
Elements	150	16^2	64^2	32^2	80×35	160×70	128^2	64^2	80×240
p4est initial refinement (levels)	—	—	yes (1)	—	—	yes (1)	yes (2)	yes (1)	yes (2)
N	4	4	4	7	4	4	3	3	4
Δt	5×10^{-5}	2×10^{-3}	2×10^{-4}	2×10^{-3}	7.5×10^{-3}	5×10^{-3}	2×10^{-4}	10^{-5}	1.25×10^{-7}
t_{end}	0.1	1	1	15	54	50	0.15	0.01	2×10^{-3}
Integrator	CK2N54	CK2N54	CK2N54	CK2N54	CK2N54	CK2N54	CK2N54	CK2N54	CK2N54
Stage limiter	none	none	none	none	FE floor	none	cons. limiter	cons. limiter	cons. limiter
CG, LGL, lumped mass, `TOTAL`, `"energy"`, fixed Δt	yes	yes	yes	yes	yes	yes	yes	yes	yes
Filter / KEP / ES / adapt	off	off	off	off	off	off	off	off	off
Boundaries	Dirichlet ends	periodic	periodic	periodic	periodic x, walls	periodic x, walls	periodic	periodic	inflow, outflow, axis
Sources	none	ψ only	ψ only	ψ only	g, WB, ψ, sponge	g, WB, ψ, sponge	ψ only	ψ only	ψ only
GLM c_h	n/a	IC max	IC max	IC max	IC max	IC max	IC max	IC max	IC+inflow max
ψ damping	n/a	$c_h/0.18$	$c_h/0.18$	$c_h/0.18$	$0.2\, c_h/\Delta h$	$0.2\, c_h/\Delta h$	$c_h/0.18$	$c_h/0.18$	$c_h/0.18$

Table 2 — DynSGS (effective values)

	BW	SV	OT	KH	FE	FED	ROT	BL	AJ
`:visc_model`	DSGS_MHD	DSGS_MHD	DSGS_MHD	**SMAG**	DSGS_MHD	DSGS_MHD	DSGS_MHD	DSGS_MHD	DSGS_MHD
`:`μ	all 1	all 1	all 1	$(0, 1, \dots)$	all 1	all 1	all 1	all 1	all 1
Sensor	residual	residual	legacy	n/a	legacy	legacy	legacy	legacy	residual
Norms	domain	domain	domain	n/a	element	element	element	element	element
Fast-speed floors	no	no	no	n/a	no	no	no	yes	yes
Thermal residual	no	no	no	n/a	no	no	no	yes	no
C_R	1*	1	1	n/a	1	1	1	1	1
$C_{\max}$	0.5*	0.5	0.5	n/a	0.5	0.5	0.5	0.5	0.5
$C_{\min}$	0.06*	0	0	n/a	0.03	0.03	0	0	0
$\Delta_K = \Delta_K^{elem}/(N+1)$ (value)	1/750	0.25	1/320	n/a	0.2	0.1	1/512	1/256	0.00125
`:dsgs_gamma`= γ, Pr = 1, `rel`=`local_rel`= 1	yes	yes	yes	n/a	yes	yes	yes	yes	yes
Hold (steps)	2	2	2	n/a	2	2	2	0	0
Cutoff, freeze, nodal ν, nodal ρ, reference	off§	off§	off	n/a	off	off	off	off	off§
`:dsgs_conserved`	true	false†	false	n/a	true	true	true	true	true
Diffused variables	$\mathbf{q}$	$\rho, u, v, T, w, \mathbf{B}, \psi$	$\rho, u, v, T, w, \mathbf{B}, \psi$	$(u, v, T, \dots)$	$\mathbf{q} - \mathbf{q}_e$	$(\mathbf{q} - \mathbf{q}_e)/\rho_e$	$\mathbf{q}$	$\mathbf{q}$	$\mathbf{q}$
Slot coefficients	ν all	ν; $\bar{\rho}\nu$ ($\mathbf{m}$, E)†	ν; $\bar{\rho}\nu$ ($\mathbf{m}$, E)	μ_t-based	ν; E_{th}: κ^E	$\nu\rho_e$; E_{th}: $\kappa^E\rho_e$	ν all	ν all	ν all
Work terms in E	none	$\tau \cdot \mathbf{v}$, $\eta\mathbf{B} \cdot \nabla\mathbf{B}$†	$\tau \cdot \mathbf{v}$, $\eta\mathbf{B} \cdot \nabla\mathbf{B}$	$\tau \cdot \mathbf{v}$	none	none	none	none	none
Mass diffusion $\nabla \cdot (\nu\nabla\rho)$	on	on	on	off	on	on	on	on	on
Residual max over	8 slots	8 (no ψ)	8 (no ψ)	n/a	8 (no ψ)	8 (no ψ)	8 (no ψ)	8 (no ψ) +ρe	8 (no ψ)
Dirichlet-node zeroing	yes (ends)	— (periodic)	— (periodic)	n/a	no (legacy path)	no (legacy path)	— (periodic)	— (periodic)	yes

* BW since `2bb4c2a1`: the kernels' $\Delta x/(N+1)$ with $(C_R, C_{\max}, C_{\min}) = (1, 0.5, 0.06)$ unscaled. Before, its deck multiplied them by $((N+1)/N)^2$, $(N+1)/N$, $(N+1)/N$ (Dao & Nazarov's $\Delta x/N$: 1.5625, 0.625, 0.075 at $N = 4$), which `JEXPRESSO_BW_HSCALE=nop` still gives. FE/FED: $\kappa^E = \max(\gamma(\gamma-1)/\Pr \cdot \nu_c,\ \nu_{\text{fl}}) = \max(0.0525\,\nu_c, \nu_{\text{fl}})$ on the thermal part $p/(\gamma-1)$ of E; the non-thermal part takes ν. OT, SV: $\bar{\rho}$ = element mean, $\kappa = \bar{\rho}\nu/\Pr$ on ∇T. † SV since `f9cdb02` (`:dsgs_conserved => false`, `:dsgs_nazarov_energy => true`); at `dcc3653` it had `:dsgs_conserved => true`: ν on every slot of $(\rho, u, v, T, w, \mathbf{B}, \psi)$ and no work terms (finding 2). § Freeze on since `b956ccd`: `:dsgs_freeze_stage` defaults to true for `DSGS_MHD` with the residual sensor (BW, SV, AJ); the legacy-sensor cases are unchanged and ignore it (finding 8).

Table 3 — Positivity

	BW	SV	OT	KH	FE	FED	ROT	BL	AJ
`:lpositivity`	false	false	false	false	false	false	true	true	true
State modified to enforce $\rho, p > 0$	never	never	never	never	case floor	never	cons. limiter	cons. limiter	cons. limiter
When	n/a	n/a	n/a	n/a	stages 2–5 + end	n/a	stages 2–5 + end	stages 2–5 + end	stages 2–5 + end
$\rho_{\min}$	n/a	n/a	n/a	n/a	$0.2\,\rho_e(z)$	n/a	10^{-6}	10^{-6}	1.4×10^{-7}
$p_{\min}$	n/a	n/a	n/a	n/a	$0.2\,p_e(z)$	n/a	10^{-6}	10^{-7}	10^{-6}
Floors relative to	n/a	n/a	n/a	n/a	reference state	n/a	$10^{-6}\times$ ambient	$10^{-6}\times$ ambient	$10^{-6}\times$ ambient
Conserves ρ, $\rho\mathbf{v}$, E	n/a	n/a	n/a	n/a	no (adds ρ, E)	n/a	yes (element pass)	yes (element pass)	yes (element pass)
Fallback	n/a	n/a	n/a	n/a	n/a	n/a	node floor	node floor	node floor
Touches $\mathbf{B}$, ψ	n/a	n/a	n/a	n/a	no	n/a	no	no	no
γ from	n/a	n/a	n/a	n/a	γ_{mhd}	n/a	`:dsgs_gamma`	`:dsgs_gamma`	`:dsgs_gamma`
Report	n/a	n/a	n/a	n/a	rank-local, per decade	n/a	every 1000 calls	every 1000 calls	every 250 calls
Flux-only floors	$p \geq 10^{-9}$	—	—	—	$p \geq 10^{-9}$	$p \geq 10^{-9}$	—	—	$\rho \geq 10^{-14}$, $p \geq 10^{-6}$

"Stages 2–5 + end": the stage limiter of `CarpenterKennedy2N54` is called before the right-hand side of stages 2–5 and once after stage 5, i.e. on every state the right-hand side sees except the initial condition. For ROT, BL, AJ this follows from the rebuilt solver's `williamson_condition = false`; for FE from the same keyword in its deck (since `e448f7d`; at `dcc3653` only from `calck = true`, the non-empty `saveat` of Jexpresso's `solve`). The node-wise *repair* of `src/kernel/positivity` (`:positivity_method => "repair"`) is used by none of the nine cases.

Common to all cases where it applies (solid underline)

- **General (9/9):** CG spectral elements on LGL nodes with a lumped mass matrix; `TOTAL()` variables with total energy in slot 4 and the slot order $(\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z[, \psi])$; ideal-gas MHD pressure; `CarpenterKennedy2N54` at fixed Δt; `:lvisc => true`; filter, KEP, entropy variables, Richardson correction and AMR off; CPU only.
- **GLM (8/8 in 2D):** $c_h = \max(|\mathbf{v}| + c_f)$ of the initial (AJ: and inflow) data, constant in time; ψ is damped by a linear source.
- **DynSGS (8/8):** `DSGS_MHD`, one coefficient per element recomputed at every RK stage (since `b956ccd` once per step for the residual-sensor cases BW, SV, AJ; finding 8), element length $\Delta_K^{elem}/(N+1)$, history rotated once per step, max over the eight non-ψ normalized residuals, cap $C_{\max}\Delta_K \max(|\mathbf{v}| + c_f)$ with c_f bounded by $\sqrt{(\gamma p + |\mathbf{B}|^2)/\rho}$ (BW: exact x-fast speed), all multipliers $\mu = 1$ (mass diffusion on), $C_R = 1$ and $C_{\max} = 0.5$ (BW since `2bb4c2a1`), `:dsgs_gamma` $= \gamma$, Pr $= 1$, `:dsgs_rel` = `:dsgs_local_rel` $= 1$, no cutoff, no stage freezing, no nodal coefficient, no nodal-ρ scaling, no reference-RHS subtraction.
- **Conservative limiter (3/3):** hook, validation, γ from `:dsgs_gamma`, floors at 10^{-6} of the ambient ρ and p, element pass that conserves ρ, $\rho\mathbf{v}$, E and leaves $\mathbf{B}$, ψ untouched, DSS of the increments, non-conservative node floor as fallback.

Shared by a majority (dashed underline)

2D with nine slots (8/9); $N = 4$ (6/9); ψ damping $c_h/0.18$ (6/8); `:lsource` on (8/9); doubly periodic (5/9); p4est initial refinement (5/9); no state modification for positivity (5/9) and `:lpositivity` false (6/9); among the DynSGS cases: legacy sensor (5/8), element norms (5/8), $C_{\min} = 0$ (5/8), hold 2 (6/8), `:dsgs_conserved` (6/8), no work terms in E (6/8), no fast floors (6/8), no thermal residual (7/8), no reference weight (7/8) (`:dsgs_nazarov_energy` is now 4 true, 4 false: no majority); report every 1000 calls (2/3 limiter cases).

Findings: code that disagrees with its comments, and one failure

Line numbers refer to `dcc3653`; fixes made since are marked with their commit.

1. [**Fixed in `23e4974`.**] **BW, element form (the default): `MethodError` on the first DynSGS call.** Commit `708bd78` added `lthermal = ...` to the 1D call (`rhs.jl:1456`) but not to `compute_dsgs_viscosity!(::DSGS_MHD, ::NSD_1D)` (`SGS.jl:1929`). Reproduced with Julia 1.11.5 on the kernel's own source ("got unsupported keyword argument `"lthermal"`"). `23e4974` adds `lthermal::Bool=false` to the 1D signature (the thermal residual is 2D-only and is ignored there); the call then succeeds with a bit-identical coefficient for `lthermal` omitted, false or true.
2. [**Fixed in `f9cdb02`.**] **SV: conserved coefficients on physical primitives.** At `dcc3653` the deck set `:dsgs_conserved => true` but `user_primitives.jl` returns $(\rho, u, v, T, w, \mathbf{B}, \psi)$, so the operator was $\nu\nabla u$, $\nu\nabla T$, ... with no ρ factor, no stress and no work terms — neither documented form. `f9cdb02` sets `:dsgs_conserved => false` and `:dsgs_nazarov_energy => true`: the physical form of Orszag–Tang that the primitives' header describes. Measured here over the 16-run grid of `tools/smooth_vortex_scan.sh` ($N = 4$–7, 4^2–32^2): the velocity L^1 error changes by -4% to $+3\%$ on the two finest meshes, with the same observed orders; 10–14% lower on 4^2, $1.21\times$ at $N = 6$, 8^2 (details in `smoothVortex.pdf`).
3. [**Fixed in `f03762b`.**] **FE, FED:** the comments that `"energy"` keeps the $\tau \cdot u$ term active, that $\Delta t = 2.5 \times 10^{-3}$, and that the energy slot is scaled by $\gamma(\gamma-1)/\,\mathrm{Pr}$ were stale; FED's header and README claimed the sibling's mesh and time step, while it refines the mesh once and uses $\Delta t = 5 \times 10^{-3}$ (FE: 80×35, 7.5×10^{-3}). Comments and READMEs now match the decks (comments only; no value changed).
4. [**Fixed in `e448f7d`.**] **FE:** the floor reached stages 2–5 only through `calck = true`; the deck did not set `williamson_condition = false` as the conservative limiter does. Now it does (1 limiter call per step against 5 without `calck`, measured on the installed OrdinaryDiffEqLowStorageRK 1.11.0; through Jexpresso's `solve`, which has `calck = true`, old and new decks give byte-identical output at $t = 1, 2\,\tau_0$).

5. **OT:** the comment that `"rank"` is the code-wide norm default is stale (it is `"domain"`); the comments on Δt (5×10^{-4}) and on 32^2 elements giving 128^2 points predate the current $\Delta t = 2 \times 10^{-4}$ and the one-level refinement.
6. [**Fixed in `5ed7f539`.**] **BW:** the header said SSPRK53, the deck uses `CarpenterKennedy2N54`; `DSGS.md` §4.7 quoted a 3% floor, the deck sets $0.06\,(N+1)/N$, its default all along. Both now say what the deck does.
7. **KH:** the mesh named by the deck (`meshes/gmsh_grids/hexa_TFI_32x32_unitsquare.msh`) is not in this checkout.
8. [**Changed in `b956ccd`.**] **All residual-sensor cases (BW, SV, AJ): the coefficient is frozen per step.** `:dsgs_freeze_stage` now defaults to true for `DSGS_MHD` with the residual sensor. Per stage, the residual's time stencil includes an RK internal stage with an $O(\Delta t)$ error, so ν never falls below $O(h^2\Delta t)$ on a smooth solution: on the MHD Alfvén wave at $N = 4$ that capped the order at 3.2, against 4.8 frozen. On BW the frozen and per-stage density errors differ by under 5% (under 3% without the floor), with the same or less excess total variation. The legacy sensor reads $-\frac{1}{2}\partial_t q$ at the frozen stage, so it ignores the option (ROT, BL, FE, FED, OT unchanged).

Appendix: code map (branch `sm/MHD`, commit `dcc3653`)

Mechanism	Where (file:line)
Deck defaults of every `:dsgs_*` key	`src/io/mod_inputs.jl:1085-1376`
Deck defaults of `:lpositivity`, floors, report	`src/io/mod_inputs.jl:456-477`
`:positivity_method`; rebuild of the RK solver	`src/io/mod_inputs.jl:828-838`
DynSGS history rotation, stage weights, hold, freeze	`src/kernel/operators/rhs.jl:927-984`
Stencil selection `_dsgs_stencil`	`rhs.jl:141`
Hold length `_dsgs_hold_steps` ($\max(2, H) + 1$ rotations)	`rhs.jl:182`
Residual right-hand side (element / legacy / Dirichlet)	`rhs.jl:188-280`; legacy fill `rhs.jl:282`; boundary zeroing `rhs.jl:313, 346`
1D `DSGS_MHD` dispatch	`rhs.jl:1398-1470`
2D `DSGS_MHD` dispatch (sets `dsgs_ref_weight`, `dsgs_split_energy`, `dsgs_nodal_rho`)	`rhs.jl:1517-1610`
2D viscous assembly `_viscous_rhs_el_2d_dsgs!`	`rhs.jl:1796`
2D weak viscous flux `_expansion_visc!` (conserved flag, ref. weight, split energy, $\tau \cdot \mathbf{v}$, resistive work)	`rhs.jl:2943-3210` (`lcons` 2959, weight 2978, split 3050, $\tau \cdot \mathbf{v}$ 3076, $\eta\,\mathbf{B} \cdot \nabla\mathbf{B}$ 3091)
1D weak viscous flux	`rhs.jl:2823`; element loop `rhs.jl:1321`
2D `DSGS_MHD` element kernel	`src/kernel/physics/SGS.jl:1598-1912`
1D `DSGS_MHD` element kernel	`SGS.jl:1929-2120`
Nodal kernels (off in every MHD deck; BW can switch it on with `JEXPRESSO_BW_NODAL=1`)	`SGS.jl:2144` (1D), `SGS.jl:2515` (2D)
Normalization floor `_dsgs_denom`, `:dsgs_rel` multiplier	`SGS.jl:676-679`; set per call in `rhs.jl:192`
Smoothness cutoff `_dsgs_cut`	`SGS.jl:703`
`SGS_diffusion(::DSGS_MHD)` (nodal-ρ switch)	`SGS.jl:547-569`
Element length Δ_K^{elem} (2D: min of corner extents)	`src/kernel/mesh/mesh.jl:6430-6457`; 1D Δx: `mesh.jl:5863`
`uprimitive` width ($n_{eqs} + 2$ in 2D, $n_{eqs} + 1$ in 1D)	`src/kernel/globalStructs.jl:139-153`
Realizability repair (`"repair"`, unused by the MHD decks)	`src/kernel/positivity/Positivity.jl:294`; driver `positivity_driver.jl:124`; called in `rhs.jl:821`
Conservative element limiter (`"conservative"`)	`Positivity.jl:433-559`; driver `positivity_driver.jl:245-337`
Main `solve` call (`saveat` non-empty ⇒ `calck = true`)	`src/kernel/solvers/TimeIntegrators.jl:855`

Stage-limiter calls in `CarpenterKennedy2N54`: with `williamson_condition = false` the limiter is called before the right-hand side of stages 2–5 and once after stage 5 (OrdinaryDiffEq v6.90/v6.95, `low_storage_rk_perform_step.jl`). In OrdinaryDiffEq `calck = true` forces `williamson_condition = false` in the 2N cache, and Jexpresso's `solve` passes a non-empty `saveat`, so this holds for every deck that hands the solver a stage limiter. (Jexpresso ships no `Manifest.toml`; the package version is whatever the environment resolves, and the check above was made against v6.90/v6.95.)